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Note from author

These notes are mostly based on the syllabus and past papers

Fair warning: they might look short (2–3 pages), but I've condensed everything down to keep the important stuff while removing the extra yap. That said, I wouldn't recommend using these as quick revision notes, but think of them as detailed notes covering almost everything

Spotted a mistake? Suggestions?

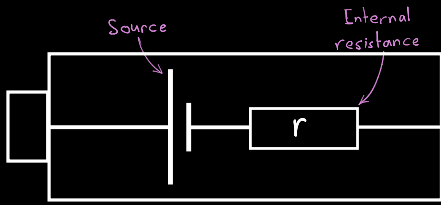
I'd love to hear your feedback. Contact us at: learnmates.share@gmail.com

circuit symbols

cell		switch	
battery of cells	 or 	earth	
power supply		electric bell	
a.c. power supply		buzzer	
junction of conductors		microphone	
lamp		loudspeaker	
fixed resistor		motor	
variable resistor		generator	
thermistor		ammeter	
light-dependent resistor		voltmeter	
heater		galvanometer	
potentiometer		oscilloscope	
diode		capacitor	
light-emitting diode		Not all symbols shown are required for AS – some appear in A2. You will become familiar with the frequently used ones as you solve past papers.	

Internal Resistance

A real battery has internal resistance (r). It behaves like an ideal battery with e.m.f. E in series with a small resistor r .
Why? Charge carriers collide with atoms inside the battery, transferring energy as heat.
Effect: Terminal p.d. provided is less than actual e.m.f. of battery
Explanation: e.m.f. is total energy available per unit charge, but some of the available volts/energy is lost in the internal resistance of the battery when it supplies current (hence p.d. available less than e.m.f.)



Terminal Potential Difference Equation

Terminal P.d. is the voltage actually supplied by the battery, and it can be calculated using:

$$V = E - I r$$

Volts lost

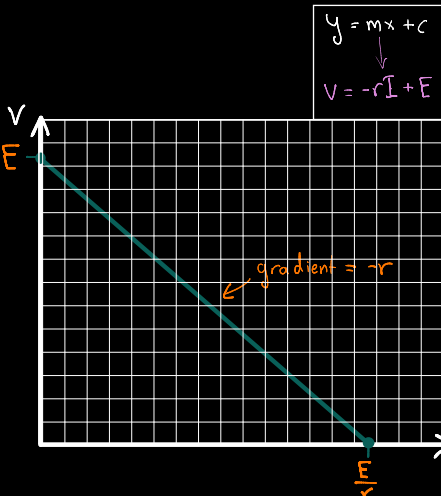
Where:

- V – terminal p.d. (V)
- E – e.m.f. (V)
- I – current (A)
- r – internal resistance (Ω)

Terminal p.d. vs Current

As current increases, terminal p.d. decreases.
Why? More current $\rightarrow$ more energy/volts lost (as heat) internally $\rightarrow$ less energy/voltage available at the terminals.

Connection with external resistance: As external resistance increases, current decreases. And using $V = E - Ir$, we can deduce that a smaller current means fewer volts are lost across internal resistance. Therefore, more of the e.m.f. is supplied as terminal p.d.
Conversely, if external resistance decreases $\rightarrow$ current increases $\rightarrow$ more volts are lost internally, and terminal p.d. decreases.



Kirchhoff's laws

Kirchhoff's First Law

Statement:

'The sum of currents into (entering) a junction = the sum of currents out of (leaving) the junction.'

Equation:

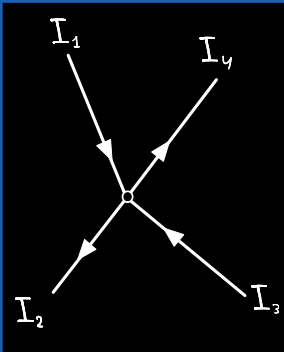
$$\sum I_{in} = \sum I_{out}$$

Physical basis: Conservation of charge. Charge cannot accumulate at a junction – whatever charge flows in must flow out.

Example 3

$$\sum I_{in} = \sum I_{out}$$

$$I_1 + I_2 = I_3 + I_4$$



Kirchhoff's First Law

Statement:

'The sum of e.m.f.s (electromotive forces) = the sum of p.d.s (potential differences) around a loop.'

Equation:

$$\sum E = \sum IR$$

Physical basis: Conservation of energy. Energy given to charges by sources (e.m.f.) equals energy lost by charges in components (p.d.).

Example 3

Use Kirchhoff's laws to state the relationship between E_1 , R_1 , I_1 and I_2 in loop BCKYB.

1) Draw loop & choose direction

2) Apply the law

– Loop direction goes from – to + so E_2 is positive

– Both I_1 and I_2 are in the same direction as loop path so they are both positive

– Equation:

$$\sum E = \sum IR \Rightarrow E_2 = I_1 R + I_2 R$$

Energy Transfers in Circuits (p.d. and e.m.f.)

Potential difference (p.d.) is energy transferred from electrical to other forms (heat, light, sound, etc.) per unit charge. It is measured across two points (component) – it's the "drop" in energy as charge goes through.

$$V = \frac{W}{Q}$$

Where:

- V – potential difference, unit: volts (V)
- W – energy transferred (J)
- Q – charge (C)

Electromotive force (e.m.f.) is the energy transferred from other forms/chemical to electrical per unit charge as charge passes through a source (battery, generator)

$$E = \frac{W}{Q}$$

Where:

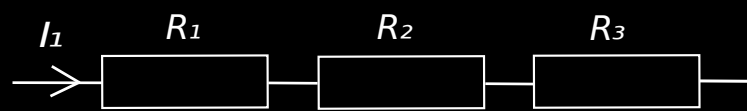
- E – e.m.f., unit: volts (V)
- W – energy transferred (J)
- Q – charge (C)

When asked to distinguish between e.m.f. and p.d.: Compare the energy conversion:

- e.m.f.: other forms $\rightarrow$ electrical
- p.d.: electrical $\rightarrow$ other forms

Resistors in parallel and series

Resistors in series



Current

$$I_T = I_1 = I_2 = I_3 = \frac{V_T}{R_T}$$

(Note: I_T is labeled as total current in the original image)

In a series circuit, charge carriers have **only one route**. The rate of flow (**current**) **cannot change** - whatever flows through one resistor must flow through all.

Voltage

$$V_T = V_1 + V_2 + V_3$$

(Note: V_T is labeled as total across arrangement in the original image)

The total p.d. supplied to arrangement is shared among the resistors. The sum of p.d.s across each resistor equals the total p.d.

Resistance

$$R_T = R_1 + R_2 + R_3$$

(Note: R_T is labeled as total in the original image)

Derivation (using Kirchhoff's law):

$$\begin{aligned} \sum E &= \sum IR \\ \Rightarrow [IR_T] &= [IR_1 + IR_2 + IR_3] \div I \quad \text{Constant current} \\ \Rightarrow R_T &= R_1 + R_2 + R_3 \end{aligned}$$

Power

$$P_T = P_1 + P_2 + P_3$$

For series circuit use $P = I^2R$, as current I is the same everywhere, so comparing power is just comparing R .

$$\Rightarrow P \propto R$$

Conclusion: The bigger the resistance, the more power it dissipates. (given current is constant ofc)

Reason: A larger resistor gets a greater share of the total voltage ($V = IR$, if I is constant, larger R means larger V , so larger $P = VI$)

Ammeter



What it measures: Current (A)

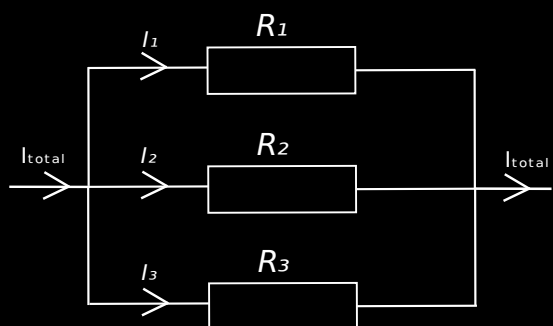
How it is connected: In series with the component

Reason: Ammeter measure current and current is constant in series

Resistance: Ammeter must have low resistance (to not affect current)

What if it is connected in parallel: current would split and most current would pass the ammeter (low resistance path) → incorrect reading, may damage ammeter

Resistors in series



Current

$$I_T = I_1 + I_2 + I_3$$

(Note: I_T is labeled as total in the original image)

At a junction, charge is conserved. The **total current entering equals the sum of currents leaving** (Kirchhoff's 1st Law)

Voltage

$$V_T = V_1 = V_2 = V_3$$

(Note: V_T is labeled as total across arrangement in the original image)

Unlike current, which splits at junctions, the voltage supplied is the **same across all branches** in a parallel circuit.

Power

$$P_T = P_1 + P_2 + P_3$$

For parallel circuit use $P = V^2/R$, as voltage V is the same everywhere, so comparing power is just comparing $1/R$.

$$\Rightarrow P \propto \frac{1}{R}$$

Conclusion: The smaller the resistance, the more power it dissipates.

Reason: A smaller resistor allows more current to flow (since $I = V/R$). More current means more charge carriers per second, each transferring energy, so total power is higher ($E = IV$, and V is constant)

Voltmeter



What it measures: Potential difference (V)

How it is connected: In parallel across the component

Reason: Voltmeter measures the difference in potential between two points (across)

Resistance: Voltmeter must have high resistance (to not draw current)

What if it is connected in series: Voltmeter would reduce the current / voltage through component as a voltmeter has very high resistance → component gets very little voltage/current, voltmeter reads nearly the full battery voltage

Why is Total Resistance Less Than the Smallest Resistor?

Adding more paths (branches) makes it easier for current to flow. More paths = lower total resistance.

Example: $R_1 = R_2 = 10\Omega$

$$R_T = \frac{10 \times 10}{10 + 10} = 5\Omega$$

$5\Omega < R$

Resistance

$$\frac{1}{R_T} = \frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3}$$

(Note: R_T is labeled as total in the original image)

Derivation (using Kirchhoff's law):

$$\begin{aligned} \sum I_m &= \sum I_{out} \Rightarrow [I_T + I_1 + I_2 + I_3] \div V \\ \Rightarrow \frac{I_T}{V} + \frac{I_1}{V} + \frac{I_2}{V} + \frac{I_3}{V} &\Rightarrow \left[\frac{1}{R_T} + \frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3} \right] \end{aligned}$$

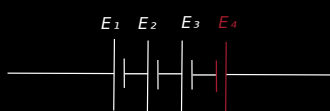
Special Case: Two Resistors in Parallel

$$\begin{aligned} \frac{1}{R_T} &= \frac{1}{R_1} + \frac{1}{R_2} \\ &= \frac{R_1 + R_2}{R_1 \times R_2} \end{aligned}$$

$$R_T = \frac{R_1 \times R_2}{R_1 + R_2}$$

Cells in parallel and series

Cells in series

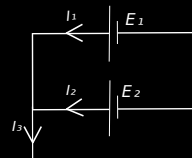


What happens: Charge passes through each cell one after another. Each cell gives energy to the charge. So total energy gained = sum of all cells.

Rule: You add them up (if direction is reversed → opposes the flow → instead of adding energy, it **subtracts energy**)

Solution to the example above: $E_i = E_1 + E_2 + E_3 + E_4$

Cells in parallel



What happens: Charge can choose which cell to go through. Each cell supplies only part of the total current. So total e.m.f. stays the same.

Rule: Current: Total current = sum of currents from each cell.

Voltage: Same as one cell with e.m.f. E

(Imp. requirement: All cells must have the same e.m.f.)

Solution to the example above: **Current:** $I_3 = I_1 + I_2$

Voltage: $E_i = E_1 = E_2$

Exam Tips

Tip 1: Comparing Two Components

When asked to compare components (e.g. power consumption), use a formula that has most variables the same for both, and only one (preferably) variable different.

Example: In a parallel circuit, voltage V is the same across all branches. Use $P = V^2/R$. Since V is constant, power is simply proportional to $1/R$. The smaller the resistance, the greater the power.

Tip 2: Effect of Changing a Variable

When asked about the effect of a change (e.g. increasing resistance, adding a resistor, temperature increase), follow these steps: ask these questions:

- 1 What happens to the resistance of the component?
- 2 What happens to the total resistance of the circuit?
- 3 What happens to the current (or total current)?
- 4 Does it affect voltage shared across components?
- 5 Use an equation (e.g. $P = IV$, $V = IR$) to reach a conclusion.

Example: A filament lamp heats up, what happens to a resistor in series? Resistance increases → total resistance increases → current decreases → power $P = I^2R$: I decreases, R of series resistor constant, so power decreases

Using $P = IV$: both I and V decrease → power decreases